

## Algorithms: Some Proofs about Trees

Model 1: A theorem about trees

**Learning objective:** Students will write proofs about graphs.

**Theorem 1** (Trees). Let  $G = (V, E)$  be a graph with  $|V| = n \geq 1$ . Any two of the following imply the third:

1.  $G$  is connected.
2.  $G$  is acyclic.
3.  $G$  has  $n - 1$  edges.

- 1 (Review) From the previous activity, what is the definition of a tree?
- 2 How does the given theorem relate to the definition of a tree?
- 3 The theorem implies that there are two other alternative, yet equivalent, ways we could have defined trees. What are they?

We will take each pair of statements in turn and show that they imply the third. Fill in the blanks to complete the following proofs! Note that the size of a blank does not necessarily correspond to the amount of stuff you should write in it.

**Lemma 2.**  $(1) \wedge (2) \implies (3)$ . That is: let  $G = (V, E)$  be a graph with

$|V| = n \geq 1$ . If \_\_\_\_\_

and \_\_\_\_\_,

then \_\_\_\_\_.

*Proof.* Let  $P(n)$  denote the statement “Any graph  $G$  with  $n$  vertices

which is \_\_\_\_\_ and \_\_\_\_\_

must have \_\_\_\_\_.”

We wish to show that  $P(n)$  holds for all  $n \geq 1$ .

The proof is by \_\_\_\_\_.

- The base case is when \_\_\_\_\_.

In this case,  $G$  must be \_\_\_\_\_

which indeed \_\_\_\_\_.

- For the induction step, suppose  $P(k)$  holds for some  $k \geq 1$ . That is,

suppose that any graph with \_\_\_\_\_ vertices

which is \_\_\_\_\_

must have \_\_\_\_\_.

Then we wish to show  $P(k+1)$ , that is, any graph with \_\_\_\_\_

vertices which is connected and acyclic must have \_\_\_\_\_.

So, let  $G$  be a graph with \_\_\_\_\_ vertices which is

\_\_\_\_\_ and \_\_\_\_\_.

We claim that  $G$  must have some vertex which is a leaf, that is, a

vertex of degree \_\_\_\_\_,

which we can show as follows:

- $G$  cannot have any vertices of degree \_\_\_\_\_

because \_\_\_\_\_.

- It also cannot be the case that every vertex of  $G$  has degree  $\geq \_$ .

If they did, then we could find a \_\_\_\_\_ by starting at any



vertex and walking along edges randomly until \_\_\_\_\_;

we would never get stuck because \_\_\_\_\_.

However, this is impossible because we assumed \_\_\_\_\_.

Hence,  $G$  must have some vertex which \_\_\_\_\_.

If we delete this vertex along with the edge adjacent to it, it results

in a graph  $G'$  with only \_\_\_\_\_ vertices;

we note that  $G'$  is still \_\_\_\_\_

because \_\_\_\_\_

and also still \_\_\_\_\_

because \_\_\_\_\_.

Hence we may apply the inductive hypothesis to conclude that  $G'$

\_\_\_\_\_. Adding the deleted vertex and edge

back to  $G'$  shows that  $G$  \_\_\_\_\_,

which is what we wanted to show.

□

Let's do one more! (The third is a homework problem.)

**Lemma 3.**  $(2) \wedge (3) \implies (1)$ , that is, \_\_\_\_\_

\_\_\_\_\_.

*Proof.* This proof uses a *counting argument*: we will show what we wish to show by counting things in multiple ways.

Let  $c$  denote the number of connected components of  $G$ . We want

to show that \_\_\_\_\_.

Number the components of  $G$  from  $1 \dots c$ , and say that component  $i$  has  $n_i$  vertices. Then

$$\sum_{i=1}^c n_i = \underline{\hspace{2cm}}$$



because \_\_\_\_\_.

Each connected component is by definition a \_\_\_\_\_ graph;

each component must also be \_\_\_\_\_  
since we assumed that  $G$  is. Hence we may apply Lemma 2 to con-

clude that component  $i$  \_\_\_\_\_.

Adding these up, the total number of edges in  $G$  is

$$|E| = \sum_{i=1}^c \text{_____} = \text{_____}$$

But we already assumed the number of edges in  $G$  is \_\_\_\_\_,

and hence \_\_\_\_\_ as desired. □

